Start

Inputs

Sentinel-2 composite
B3, B4, B8, B11

Lidar DEM
1 m, NAVD88

MHW elevation
NOAA station / VDatum

Study area
(optional)

A Spectral classification

Compute MNDWI (B3, B11) and NDVI (B4, B8) on one 10 m reference grid

Gaussian smoothing (σ = 1 px) and Otsu threshold → spectral water / land

B Lidar preparation

DEM supplied?

No

Yes

Clip, reproject, convert to metres; aggregate 1 m → 10 m (mean, min, max); flag flat surfaces

C Elevation zones relative to MHW (with DEM)

Lower elevation by vegetation offset (0.15 m) where NDVI > 0.3

Elevation > MHW + 0.5 m?

Yes → **Land**

No

Elevation < MHW − 0.5 m, or flat and unvegetated?

Yes → **Water**

No

NDVI > marsh threshold?

Yes → **Marsh**

No

Keep MNDWI class

Fuse with spectral result; morphology; re-apply zones

D Connectivity, tracing and shoreline-type labelling

With DEM

Connect water to the open coast through lidar channel pixels; levees block; unconnected water → inland water

Segmentation map
(marsh class needs DEM)

Without DEM (MNDWI only)

Morphology only

Open water = largest water body touching the area edge

Trace open-water boundary into segments; remove study-area border and NoData-edge segments

Marsh mask available? (DEM + Red and NIR)

No → **Shoreline, all types** unlabelled

Yes

≥ 50% of land within 2.5 px is marsh?

Yes → **Marsh shoreline** separate layer

No

Non-marsh shoreline
main output

End

Start / end

Data input / output

Process

Class assigned

Decision

All outputs projected to the output CRS; run parameters written to metadata (JSON)